

PRAMODH DAMARSHETTY

SENIOR DATA ENGINEER

CONTACT

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Stone Mountain, GA

SKILLS

- **Big Data & Streaming:** Apache Spark (Scala, Java, PySpark), Kafka, Hadoop (Hive, MapReduce), Apache Airflow
- **AWS:** Glue, Redshift, EMR, Athena, Lambda, S3, Step Functions, Kinesis, DynamoDB, RDS, IAM, CloudWatch, CloudFormation.
- **GCP:** BigQuery, Dataflow, Pub/Sub, Dataform, Data Fusion, Cloud Storage, Cloud Composer.
- **Azure:** ADF, Synapse, Databricks, Azure Functions, Cosmos DB, Azure Blob Storage, Azure Monitor
- **Snowflake** (DBT, Snowpark, Streams & Tasks), Informatica, Talend, SQL Server, PostgreSQL, MongoDB
- Jenkins, Docker, Kubernetes, Terraform, Git, CI/CD Pipelines
- RBAC, IAM Policies, Encryption, Data Governance (GDPR, HIPAA)

EDUCATION

Master of Science, Computer Science

Auburn University,
Montgomery, AL

Aug 2022 - May 2024

Bachelor of Technology, Electronics and
Communication Engineering

Sreyas Institute of
Engineering and Technology,
India

Apr 2010 - May 2014

PROFILE

Data Engineering professional with over 8 years of experience specializing in scalable data pipeline architecture, cloud data warehousing, and big data technologies. Proven track record in designing end-to-end ETL processes, executing cloud migrations, and implementing real-time data processing solutions on AWS and Azure platforms. Demonstrated ability to collaborate effectively with global clients and lead cross-functional teams to deliver robust, production-grade data solutions.

WORK EXPERIENCE

Senior Data Engineer

Capital One, Plano, TX

Sept 2023 - Present

Key Responsibilities:

- Managed and optimized large-scale distributed data processing on AWS EMR, improving Apache Spark and Hadoop job performance to enhance efficiency and reduce execution time.
- Designed and implemented ETL pipelines using AWS Glue, Apache Airflow, and AWS Lambda, ensuring seamless transformation and ingestion of structured and semi-structured data. Utilized AWS Glue bookmarks and crawlers to automate and optimize data workflows.
- Developed real-time data pipelines using Amazon Kinesis, Apache Kafka, and AWS Glue, enabling low-latency, high-throughput data ingestion and transformation for business-critical analytics.
- Designed and optimized data warehouses using Amazon Redshift, implementing partitioning, compression, and clustering strategies to improve query performance for large-scale datasets. Led on-premise ETL migrations to AWS and Snowflake, modernizing data pipelines for better scalability and cost efficiency.
- Performed advanced SQL performance tuning in Snowflake, leveraging query execution plan analysis and micro-partition pruning to optimize execution time. Designed efficient database structures using clustering keys and partitioning to enhance performance and reduce storage costs.
- Leveraged AWS DMS (Database Migration Service) to facilitate seamless cloud database migrations, ensuring minimal downtime and data integrity.
- Integrated machine learning models into an MLOps framework, ensuring scalability and efficient deployment within real-time data pipelines and AWS services.
- Developed cost-effective data lakes on Amazon S3 with AWS Glue, enabling self-service analytics and accelerating insights for business users.
- Built serverless and event-driven data pipelines using AWS Step Functions, Amazon S3, and AWS Lambda, ensuring scalable and cost-effective data processing.
- Extensive hands-on experience with PySpark, Spark with Scala and Java on Amazon EMR, leveraging Hadoop with Java for large-scale distributed data processing. Worked with Apache Sqoop, Python JDBC applications, Pandas, NumPy, Spark connectors, Fivetran, and Iceberg to streamline data ingestion.
- Designed and implemented end-to-end workflows using Apache Airflow, integrating with AWS services and Snowflake for efficient data orchestration and automation.
- Developed data transformation and modeling solutions using DBT (Data Build Tool), ensuring streamlined ETL processes and high-quality data models.

DATA ENGINEER**Key Responsibilities:**

- Designed and developed scalable data pipelines using Azure Data Factory and Apache Spark, integrating healthcare data from on-premises MySQL and Cassandra with Azure Blob Storage and Azure SQL Database. Transformed and loaded data into Azure Synapse Analytics to support enterprise-wide healthcare reporting and analytics.
- Built scalable data ingestion and transformation pipelines using Azure HDInsight, Azure Databricks, and Spark SQL, processing large healthcare datasets from Azure Data Lake Storage (ADLS Gen2) and Cosmos DB (SQL API and Mongo API) to provide real-time insights into patient care and operational efficiencies.
- Developed and optimized ETL workflows using Azure Data Factory (ADF), Spark SQL, and Python, streamlining data extraction, transformation, and loading processes into Azure Data Storage services. Automated data pipelines and workflows using Azure Logic Apps, triggering ADF pipelines and batch jobs to reduce manual intervention and improve data flow efficiency.
- Engineered real-time and batch data streaming solutions using Azure Event Hubs, Apache Kafka, Azure Synapse Analytics, and Databricks Spark Streaming. Performed RDD transformations on healthcare sensor data, enabling predictive analytics and enhancing patient care operations.
- Led large-scale SQL database migrations to Azure Data Lake Storage using Azure Data Factory, Azure SQL Database, and Azure Databricks, modernizing healthcare data infrastructure for efficient, cloud-native data management.
- Designed and optimized modern data warehouse solutions in Azure Synapse Analytics, enabling scalable, high-performance querying of structured and semi-structured healthcare data.
- Implemented serverless SQL pool and dedicated SQL pool features in Azure Synapse to reduce query latency and support near real-time analytics.
- Optimized Azure Synapse performance using resource class adjustments, partitioning strategies, and materialized views, significantly improving query execution speed on large datasets.
- Implemented robust data security and governance using Azure Key Vault, Azure Synapse, and Snowflake RBAC policies, ensuring secure access control to sensitive healthcare data and maintaining compliance with regulatory requirements.
- Managed multiple GitHub repositories to ensure efficient version control of data pipelines, ETL processes, and automation scripts, enabling collaboration, consistency, and scalability in healthcare data engineering projects.
- Created interactive dashboards and reports using Power BI integrated with Azure Synapse Analytics, providing actionable insights for healthcare operations and patient care optimization.
- Leveraged Azure Synapse Pipelines to orchestrate complex data workflows, integrating disparate datasets and improving analytics delivery speed for real-time decision-making.

CAMBIA HEALTH, PORTLAND, OR

SEPT 2021 - JUL 2022

DATA ENGINEER**Key Responsibilities:**

- Developed end-to-end data pipelines using Azure Data Factory and Snowflake to integrate data from MySQL and Cassandra with Azure Blob Storage and Azure SQL Database, transforming and loading data into Databricks Delta tables.
- Built data ingestion and transformation pipelines on Azure Databricks, using Spark SQL and Python to process data from Azure Data Lake Storage (ADLS Gen2) and Cosmos DB.
- Wrote C# code for APIs that connected on-premises systems with Azure Data Factory pipelines, automating data ingestion and validation before storing in Azure Data Lake Storage.
- Implemented batch data processing using Azure Event Hubs, Azure Synapse Analytics, and Databricks, performing Spark transformations on vehicle sensor data.
- Set up automated workflows using Azure Logic Apps to trigger ADF pipelines and batch jobs, reducing manual intervention in data processing.
- Participated in data migration projects to Azure Databricks and Snowflake, configuring clusters and creating ADF pipelines for data integration.
- Monitored Spark cluster performance using Azure Log Analytics, helping transition log storage from Cassandra to Azure SQL Data Warehouse.
- Applied data security configurations using Azure Key Vault and Snowflake RBAC policies to manage data access.
- Created ETL pipelines to move data from Azure Data Lake into Azure Synapse Analytics using Python & PySpark scripts.
- Built reports and visualizations in Power BI and SQL Server Reporting Services (SSRS) for business metrics and operational analytics.

AMAZON.COM AND SUBSIDIARIES

JUL 2019 - AUG 2021

DATA ENGINEER**Key Responsibilities:**

- Developed robust end-to-end ETL pipelines using Python, SQL, and PySpark, leveraging AWS cloud services such as AWS EMR, AWS Glue, EC2, and Lambda to enhance data analysis and reporting capabilities.
- Enhanced processing efficiency and achieved performance benchmarks by fine-tuning Hadoop and Spark configurations within EMR clusters.
- Integrated Snowpark functionality within the Snowflake environment, significantly improving data processing efficiency and analytical capabilities.
- Installed and configured Apache Airflow for AWS S3, creating Directed Acyclic Graphs (DAGs) to automate workflows effectively.
- Automated data ingestion processes using PySpark and Scala from diverse sources, including APIs, Amazon S3, Teradata, and Snowflake, developing over 10 PySpark scripts to streamline ETL/ELT operations.
- Executed data ingestion procedures with Sqoop to facilitate seamless data transfer from MongoDB to Hadoop, optimizing data integration and synchronization across systems to enhance processing and analysis workflows.
- Collaborated closely with data scientists and analysts to evaluate data models, enhancing predictive accuracy for machine learning algorithms.
- Optimized Amazon Redshift performance by implementing advanced distribution and sort keys, enabling faster query execution and improved reporting on large datasets.
- Migrated complex data warehouses from on-premises systems to Amazon Redshift, leveraging Redshift Spectrum for querying data stored in Amazon S3 without requiring additional data loading, significantly reducing ETL overhead.
- Built efficient ELT pipelines that utilized Amazon Redshift stored procedures and materialized views to streamline data transformations and reduce query processing times.
- Developed secure data-sharing mechanisms in Redshift, enabling seamless collaboration between teams while maintaining data privacy and governance compliance.
- Conducted performance tuning in Amazon Redshift by optimizing workloads with workload management (WLM) configurations, achieving significant query speed improvements for large-scale analytics tasks.

DATA ENGINEER

Key Responsibilities:

- Developed and maintained data pipelines utilizing Apache Spark and Hadoop technologies to process and integrate large-scale datasets from MySQL and Cassandra sources
- Engineered data ingestion and transformation workflows with Spark SQL and Python, handling both structured and unstructured data within HDFS to support enterprise analytics initiatives
- Implemented automated real-time and batch data processing solutions using Apache Kafka and Spark Streaming to enable predictive maintenance capabilities and improve operational efficiency
- Utilized various components of the Hadoop ecosystem, including Hive and HBase, for effective management and querying of large datasets to support analytics and reporting requirements
- Executed comprehensive data validation, cleaning, and optimization procedures to ensure high data quality and maximize pipeline performance